CSA6303 - THREAT INTELLIGENCE AND NETWORK SECUR

Assessment Tool 1: CONCEPT MAPPING

Faculty : Dr.V.Parthipan **SLOT A**

Topics: Network Security Fundamentals | Security Goals (CIA Triad) | Types of Network Threats |Introduction to Threat Intelligence Types of Threat Intelligence (Strategic, Tactical, perational) — Threat Intelligence Lifecycle — Applications in IT Industry

CO1: Interpret security principles, network threats, and the role of threat intelligence in cybersecurity.

Total: 50

Student Name : G. Aswartha Manidheep **Register number: 192425089**

Code of Conduct:

I G. Aswartha Manidheep (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

CO1 – Concept Mapping Questions

No.	Concept-Mapping Question
1	**Construct a comprehensive concept map explaining the foundations of Network Security.** Connect network security fundamentals with security objectives, the **CIA Triad (Confidentiality, Integrity, and Availability)**, network assets, vulnerabilities, threats, and protective measures, and explain how these concepts collectively support secure network operations.
2	**Develop a concept map illustrating the CIA Triad as the foundation of network security.** Relate Confidentiality, Integrity, and Availability to different types of network threats and demonstrate how a single threat can affect one or more security goals.
3	**Construct a concept map classifying different types of Network Threats and their potential impact on security goals.** Connect threats such as malware, phishing, denial-of-service, spoofing, eavesdropping, and unauthorized access with the CIA Triad and explain the resulting security implications.
4	**Develop an integrated concept map connecting Network Security Fundamentals, Network Threats, and Threat Intelligence.** Show how identifying assets, vulnerabilities, and threats leads to the collection and analysis of threat information and ultimately supports informed security decisions.
5	**Construct a concept map explaining the concept of Threat Intelligence and its role in modern network security.** Relate threat actors, indicators, vulnerabilities, attack patterns, threat information, analysis, and actionable intelligence to demonstrate how Threat Intelligence helps organizations anticipate and respond to network threats.
6	**Develop a comparative concept map of Strategic, Tactical, and Operational Threat Intelligence.** Connect each type with its purpose, target audience, time horizon, information requirements, decision-making process, and appropriate network-security use cases, and analyze how the three intelligence types complement one another.
7	**Construct a concept map representing the complete Threat Intelligence Lifecycle.** Connect planning/direction, collection, processing, analysis, dissemination, and feedback with the identification of network threats and security requirements, showing how feedback continuously improves the intelligence process.
8	**Develop an end-to-end concept map showing how a network threat progresses from identification to actionable Threat Intelligence.** Connect network security fundamentals, CIA security goals, threat detection, intelligence collection, processing, analysis, dissemination, and response, and explain the relationship between each stage.
9	**Create a scenario-based concept map for an organization experiencing a major network attack.** Integrate the CIA Triad, types of network threats, Strategic Intelligence, Tactical Intelligence, Operational Intelligence, and the Threat Intelligence Lifecycle to demonstrate how the organization can understand, prioritize, and respond to the threat.
10	**Design a comprehensive concept map demonstrating the applications of Network Security and Threat Intelligence in the IT industry.** Connect network security fundamentals, CIA Triad, network threats, Threat Intelligence, Strategic/Tactical/Operational Intelligence, and the Threat Intelligence Lifecycle with

No.	Concept-Mapping Question
	applications such as threat monitoring, incident response, vulnerability management, risk assessment, security operations, and organizational decision-making.

Register Number	Name of the Students	CO1 AT1 Questions
192465054	AJJUGUTTU ABHINAVA REDDY	Q1 & Q2
192365035	AMMINENI SUCHARITHA	Q3 & Q4
192465061	ANNAMREDDY BHARGAV REDDY	Q5 & Q6
192465033	BOMMISETTY VENKATESWARALU	Q7 & Q8
192311098	BOTCHA HARI SAI	Q9 & Q10
192465058	DUSYAANT R	Q1 & Q2
192521124	Davis Damien M	Q3 & Q4
192365038	EPURU DHARMIK REDDY	Q5 & Q6
192425089	GUNDA ASWARTHA MANIDHEEP	Q7 & Q8
192412390	HARIHARAN G B	Q9 & Q10
192311096	K MUNI KULADEEP	Q1 & Q2
192411156	KEERTHIPATI HEMA SREE	Q3 & Q4
192465057	KEVINASHREE V K	Q5 & Q6
192465005	KHADEER BASHA M	Q7 & Q8
192365025	KONGARA SAI	Q9 & Q10
192365017	LINGALA MITHUN KUMAR REDDY	Q1 & Q2
192311133	MARINENI VARUNKUMAR	Q3 & Q4
192365032	Mokharala Devi Vara Prasad	Q5 & Q6
192311129	NARRA HARSHITH	Q7 & Q8
192365053	NIMMALA CHAWRAN ARYAN	Q9 & Q10
192465051	P P ANKAIAHGARI LAKSHMI SRINIVASULU REDDY	Q1 & Q2
192365052	PALURU LIKITH VARDHAN	Q3 & Q4
192311128	PATAKOTTU VENKATA PRANAVA SESHA SAI	Q5 & Q6
192511446	Pavankumar C	Q7 & Q8
192465027	RESHMA S	Q9 & Q10

192465008	S TANISHA	Q1 & Q2
192465018	Surya V	Q3 & Q4
192465038	THAMMISETTI RAJU	Q5 & Q6
192465017	THARUN ADITH S	Q7 & Q8
192465052	V RAKSHMITHA	Q9 & Q10
192411211	V TEJA SREE	Q1 & Q2
192311265	VEMIREDDY VIKAS REDDY	Q3 & Q4

Recommended Bloom's Levels

- **Analyze (L4):** Q1, Q2, Q3, Q4, Q5, Q7, Q8
- **Evaluate (L5):** Q6, Q9
- **Create (L6):** Q10

These questions are deliberately framed so that students must **connect concepts in a hierarchy/relationship map**, rather than simply reproduce definitions.

ANSWERS

CO1 - Concept Mapping: Questions 7 and 8

Question 7: The Complete Threat Intelligence Lifecycle

Question: *Construct a concept map representing the complete Threat Intelligence Lifecycle. Connect planning/direction, collection, processing, analysis, dissemination, and feedback with the identification of network threats and security requirements, showing how feedback continuously improves the intelligence process.*

7.1 Overview

The Threat Intelligence Lifecycle is a repeatable, six-phase process that converts raw, scattered data about cyber threats into finished, actionable intelligence that decision-makers can use. It is cyclical rather than linear: the output of the last phase (feedback) becomes the input of the first phase (planning), so the process becomes more accurate, more relevant and faster with every cycle. The lifecycle is driven by two things on the left of the map: the network threats an organization has identified (malware, phishing, denial-of-service, spoofing, eavesdropping and unauthorized access) and its security requirements (the CIA Triad goals, its critical assets and its known vulnerabilities). Together these tell the intelligence team what it needs to know.

7.2 Concept Map

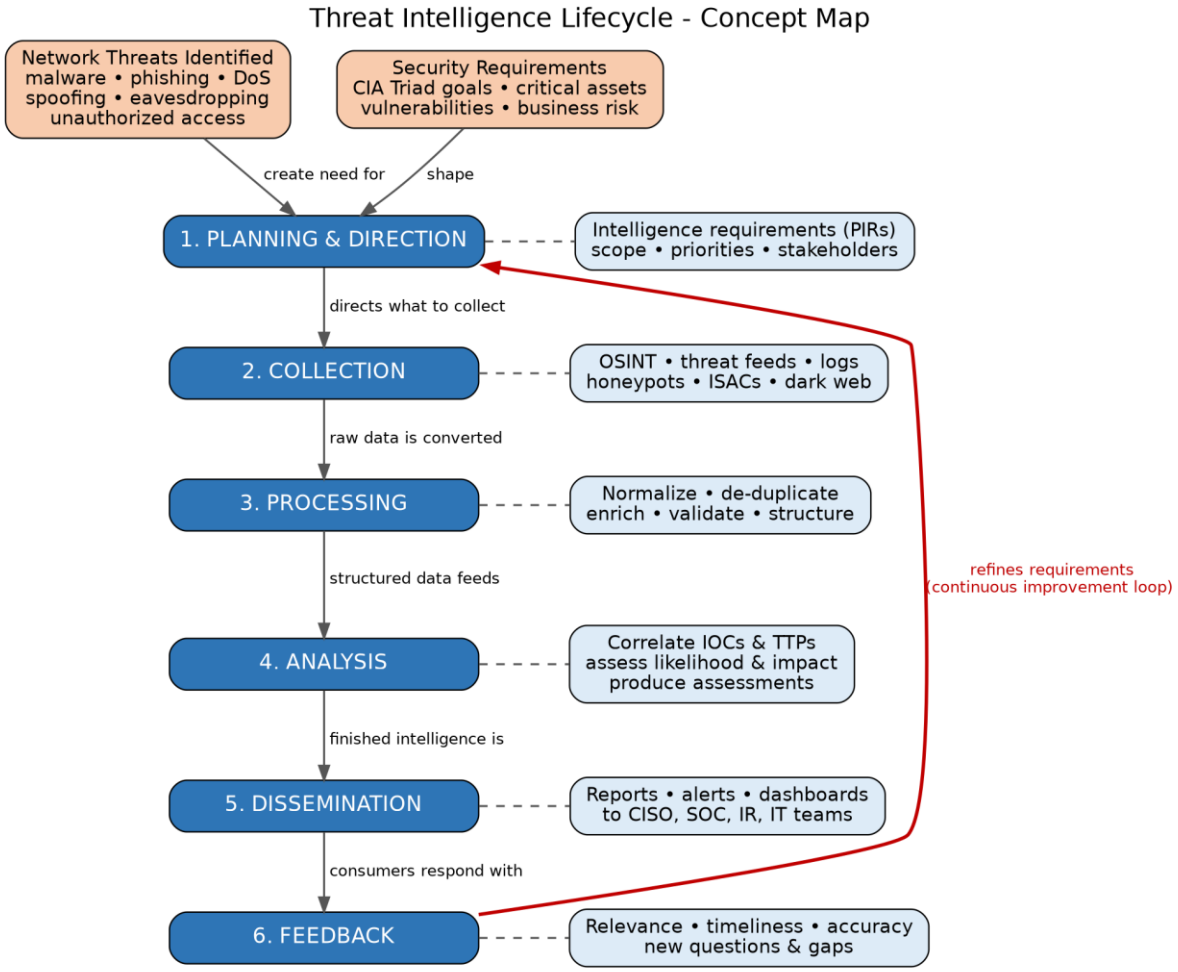


Figure 1: Concept map of the Threat Intelligence Lifecycle, linked to network threats and security requirements.

7.3 How Network Threats and Security Requirements Drive the Lifecycle

- **Network threats identified:** Each known threat type raises a question that needs an answer. For example, a rise in phishing e-mails raises the question "Which actor is behind this campaign and what malware does it deliver?" These questions become the intelligence requirements.
- **Security requirements:** Confidentiality, Integrity and Availability tell the team which assets matter most (customer database, payment systems, domain controllers). Intelligence effort is focused on threats that can damage those assets, rather than on every threat in the world.
- **Vulnerabilities:** Known weaknesses (unpatched servers, weak credentials, exposed services) determine which attacker techniques are actually relevant to the organization, which narrows collection and analysis.

7.4 The Six Phases Explained

Phase 1 - Planning and Direction

This is the starting point where the requirements for the whole cycle are set. Stakeholders such as the CISO, SOC manager and IT leaders state what they need to know, and the intelligence team converts this into Priority Intelligence Requirements (PIRs), decides the scope, the deadline, the sources to be used and who will receive the result. Good planning links every later activity to a real security need, which prevents wasted effort and "data overload".

Link to network threats: the identified threat (for example, a ransomware group targeting the industry) and the organization's critical assets decide the PIRs.

Phase 2 - Collection

The team gathers raw data that can answer the PIRs. Sources include open-source intelligence (OSINT) such as security blogs and vulnerability databases, commercial and community threat feeds, industry sharing groups (ISACs), internal telemetry (firewall, IDS/IPS, DNS, proxy and endpoint logs), malware sandboxes, honeypots and, where permitted, dark-web monitoring. The aim is breadth with relevance: the data should cover the threats named in the plan.

Link to network threats: collection searches for indicators of compromise (IOCs), such as malicious IP addresses, domains and file hashes, and for attacker behaviour tied to the identified threats.

Phase 3 - Processing

Raw data is usually large, duplicated, inconsistent and in different formats. Processing makes it usable: normalizing formats (for example STIX/TAXII, JSON, CSV), removing duplicates, translating foreign-language content, decrypting or decoding data, enriching IOCs with context (geolocation, WHOIS, reputation scores) and filtering false positives. The output is structured, trustworthy information that a machine or an analyst can work with quickly.

Link to security requirements: processing tags each item with the asset, system or CIA goal it relates to, which prepares it for risk-based analysis.

Phase 4 - Analysis

Analysts turn information into intelligence by asking "so what?". They correlate external IOCs and attacker TTPs (tactics, techniques and procedures) with internal logs, check whether the organization has been or could be hit, assess the actor's capability and intent, estimate likelihood and impact, and write assessments with a confidence level. Structured techniques (for example the MITRE ATT&CK mapping, the Diamond Model and the Cyber Kill Chain) are used to reduce bias.

Link to network threats and security requirements: analysis decides which threats actually endanger which assets and which CIA goals, producing a prioritized view of risk.

Phase 5 - Dissemination

Finished intelligence is delivered to the right people in the right format at the right time. Executives receive short strategic briefings on risk and trends; SOC and incident-response leads receive operational reports on active campaigns; analysts and engineers receive tactical feeds of IOCs and detection rules that go

straight into the SIEM, firewall and EDR. Delivery can be by reports, alerts, dashboards or automated feeds, and it must be timely, clear and actionable.

Link to security requirements: each audience gets the part of the intelligence that supports its own decisions, which protects the organization's CIA goals.

Phase 6 - Feedback

Consumers of the intelligence review whether it was relevant, accurate, timely and easy to act on, and say what is still missing. This review is collected through meetings, surveys and metrics (for example how many alerts were confirmed true, how long it took to deliver intelligence, how many PIRs were fully answered). The conclusions are used to change the requirements, sources, tools and report formats.

Link to the loop: feedback re-enters Phase 1, so new threats, new assets and new gaps are reflected in the next cycle.

7.5 Summary Table of the Lifecycle

Phase	Main Question	Key Activities	Output
1. Planning & Direction	What do we need to know, and why?	Set PIRs, scope, priorities, stakeholders	Intelligence requirements and collection plan
2. Collection	Where can we get the data?	OSINT, feeds, logs, honeypots, ISACs	Raw data and raw reports
3. Processing	Is the data usable and reliable?	Normalize, de-duplicate, enrich, validate	Structured, trusted information
4. Analysis	What does it mean for us?	Correlate IOCs/TTPs, assess likelihood and impact	Intelligence assessments with confidence levels
5. Dissemination	Who needs it, and how?	Reports, alerts, dashboards, automated feeds	Actionable intelligence delivered to consumers
6. Feedback	Did it help, and what is missing?	Reviews, metrics, new requests	Refined requirements for the next cycle

7.6 How Feedback Continuously Improves the Process

The red arrow in the concept map shows the most important idea in the lifecycle: the process never ends. Each time feedback returns to the planning phase, three improvements happen. First, the requirements become sharper, because stakeholders explain what was useful and what was noise. Second, collection becomes better targeted, because poor or redundant sources are dropped and new sources are added to fill gaps. Third, analysis and delivery become faster and clearer, because report formats, automation rules and detection content are tuned. Over several cycles the organization measurably reduces false positives, shortens the time between a new threat appearing and a defence being in place, and aligns intelligence closely with business risk.

7.7 Worked Example

Scenario: the CISO asks, "Are we at risk from the ransomware group currently targeting our industry?" (Planning) The request is logged as a requirement with a one-week deadline and the customer database and file servers named as critical assets. (Collection) Analysts pull the group's known TTPs and IOCs from

OSINT, vendor feeds and an ISAC. (Processing) The IOCs are normalized, duplicates are removed and each is enriched with reputation and first-seen data. (Analysis) The IOCs are correlated with internal firewall and DNS logs; two connections to a known command-and-control domain are found and an unpatched VPN server is identified as the likely entry point. The risk to Confidentiality and Availability is assessed as high. (Dissemination) A strategic summary goes to the CISO, an operational report goes to the incident-response lead and a list of IOCs is pushed to the SIEM and firewall. (Feedback) The CISO asks which backups are protected and how quickly the VPN can be patched, and these questions become requirements for the next cycle.

7.8 Conclusion

The Threat Intelligence Lifecycle links identified network threats and security requirements to concrete security decisions through six connected phases. Planning aims the effort, collection and processing supply reliable information, analysis gives it meaning, dissemination puts it in the hands of decision-makers and feedback closes the loop. Because of this loop, the intelligence process improves continuously and stays aligned with the organization's changing threats and assets.

Question 8: From Network Threat Identification to Actionable Threat Intelligence (End-to-End)

Question: *Develop an end-to-end concept map showing how a network threat progresses from identification to actionable Threat Intelligence. Connect network security fundamentals, CIA security goals, threat detection, intelligence collection, processing, analysis, dissemination, and response, and explain the relationship between each stage.*

8.1 Overview

A network threat does not become useful intelligence by itself. It travels through a chain of connected stages. Network security fundamentals define what must be protected and how; the CIA security goals describe what a successful attack would damage; threat detection notices that something is wrong; and the intelligence stages (collection, processing, analysis and dissemination) turn the detected activity into knowledge that supports a response. The response then feeds lessons back into the fundamentals, so the organization is better protected against the next threat. The concept map below shows this end-to-end flow, with the relationship between each pair of stages written on the arrows.

8.2 Concept Map

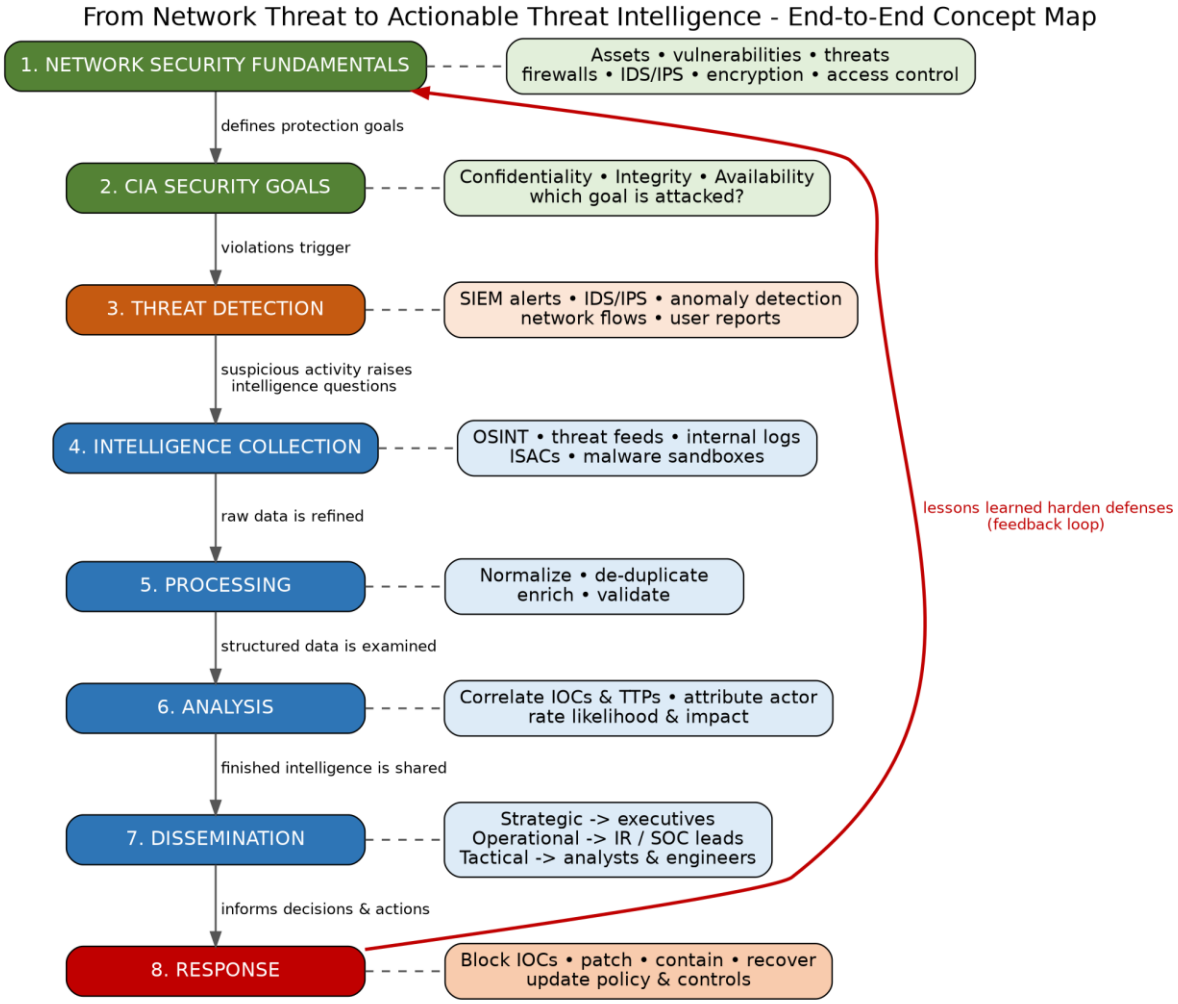


Figure 2: End-to-end concept map from network threat identification to actionable Threat Intelligence and response.

8.3 Stage-by-Stage Explanation and Relationships

Stage 1 - Network Security Fundamentals

This stage establishes the foundation: which assets exist (servers, databases, endpoints, applications, user accounts), which vulnerabilities they have, which threats could exploit them, and which controls protect them (firewalls, IDS/IPS, encryption, authentication, access control, segmentation, patching). Risk is the combination of asset value, vulnerability and threat. Without this baseline an organization cannot judge what is normal, what matters most or what an alert means.

Relationship to Stage 2: the fundamentals define the protection goals. Security controls are chosen in order to keep the assets confidential, correct and available.

Stage 2 - CIA Security Goals

Confidentiality means only authorized people can read the data; Integrity means data and systems are changed only by authorized parties and in authorized ways; Availability means systems and data can be

used when needed. Every network threat is understood by asking which of these goals it attacks. Eavesdropping attacks Confidentiality, spoofing and data tampering attack Integrity, denial-of-service attacks Availability, while malware, ransomware and unauthorized access can damage two or all three goals at once.

Relationship to Stage 3: a violation (or attempted violation) of a CIA goal produces observable signs, and detection is the process of noticing those signs.

Stage 3 - Threat Detection

Detection tools and people watch for signs that a goal is under attack: SIEM correlation alerts, IDS/IPS signatures, anomaly detection on network flows, endpoint alerts, unusual logins, spikes in traffic and reports from users (for example a suspicious e-mail). Detection answers "something is happening", but usually not who, how or how far it goes. Initial triage assigns a severity based on the asset and the CIA goal affected.

Relationship to Stage 4: a detection creates intelligence questions (Who is this? Is it part of a campaign? What else will they do?) that cannot be answered from the alert alone. This is the moment where the threat enters the intelligence process, and the questions act as the intelligence requirements.

Stage 4 - Intelligence Collection

The team gathers data to answer the questions raised by detection: internal logs and packet captures around the incident, malware samples run in a sandbox, OSINT and vulnerability databases, commercial and community threat feeds, ISAC reports and, where relevant, dark-web or social-media monitoring. The data collected is mostly raw IOCs (IP addresses, domains, URLs, file hashes, e-mail headers) and descriptions of attacker behaviour.

Relationship to Stage 5: collection produces large volumes of unstructured, duplicated and sometimes false data, so it must be refined before anyone can analyse it.

Stage 5 - Processing

Raw data is normalized into common formats (for example STIX/TAXII), de-duplicated, decoded, translated, validated and enriched with context such as geolocation, domain age, reputation and first/last-seen times. Low-quality or irrelevant indicators are removed. Each item is mapped to the affected assets and CIA goals.

Relationship to Stage 6: processing gives analysts clean, structured and trustworthy information so they can spend their time on judgement rather than data cleaning.

Stage 6 - Analysis

Analysts correlate the processed indicators with internal logs and assets, identify the attacker's TTPs (using frameworks such as MITRE ATT&CK, the Cyber Kill Chain and the Diamond Model), attribute the activity to a likely actor or campaign with a confidence level, and determine the scope of compromise. They estimate the likelihood and impact of the threat, predict what the attacker is likely to do next and recommend actions. At this point information becomes intelligence.

Relationship to Stage 7: the assessment is useful only if it reaches people who can act on it, in a form they can use.

Stage 7 - Dissemination

Finished intelligence is tailored to its audience. Strategic intelligence (trends, threat actors, business risk) goes to executives and the board to support budget and policy decisions. Operational intelligence (details of an ongoing campaign, attacker intent and timing) goes to incident-response and SOC leads to guide how they hunt and respond. Tactical intelligence (IOCs, signatures, detection rules, TTP details) goes to analysts and security engineers, often through automated feeds into the SIEM, firewall, IPS and EDR.

Relationship to Stage 8: dissemination is what makes the intelligence actionable, because it provides the specific facts and recommendations the responders need.

Stage 8 - Response

Using the intelligence, the organization acts. Tactical actions include blocking malicious IPs and domains, isolating infected hosts, deploying detection rules and patching the exploited vulnerability. Operational actions include coordinating containment, eradication and recovery according to the incident-response plan, and hunting for related activity. Strategic actions include changing policy, investing in controls, training staff and revising the risk assessment.

Relationship back to Stage 1 (feedback loop): lessons learned from the response (what worked, what was missed, which controls failed) are used to harden the network, update the asset and risk register and improve detection and intelligence requirements. The red arrow in the map shows that the process is a continuous cycle.

8.4 Relationship Summary Table

From stage	Relationship (arrow label)	To stage	Why the link matters
1. Network security fundamentals	defines protection goals	2. CIA security goals	Controls exist to protect confidentiality, integrity and availability of identified assets.
2. CIA security goals	violations trigger	3. Threat detection	An attack on a CIA goal creates observable evidence that tools and analysts can detect.
3. Threat detection	raises intelligence questions	4. Collection	Alerts say that something is wrong; intelligence is needed to say who, how and what next.
4. Collection	raw data is refined	5. Processing	Raw data is noisy and in mixed formats; it must be made reliable and structured.
5. Processing	structured data is examined	6. Analysis	Clean data allows correlation and judgement instead of manual data clean-up.
6. Analysis	finished intelligence is shared	7. Dissemination	Findings only have value when the right people receive them in a usable format.

From stage	Relationship (arrow label)	To stage	Why the link matters
7. Dissemination	informs decisions and actions	8. Response	Responders and executives act on tailored tactical, operational and strategic intelligence.
8. Response	lessons learned harden defences	1. Network security fundamentals	Feedback improves controls, risk assessment and future requirements (continuous loop).

8.5 How One Threat Affects the CIA Goals

Threat	Confidentiality	Integrity	Availability	Typical detection source
Malware / ransomware	Data theft before encryption	Files modified or encrypted	Systems unusable	EDR, antivirus, file-integrity alerts
Phishing	Credentials stolen	Account misuse	Account lockouts	E-mail gateway, user reports
Denial-of-service	-	-	Service unavailable	Traffic spikes, IPS, load-balancer logs
Spoofing (ARP/DNS/IP)	Traffic intercepted	Traffic and data altered	Traffic misdirected	IDS, ARP/DNS monitoring
Eavesdropping / sniffing	Data read in transit	-	-	Hard to detect; prevented by encryption
Unauthorized access	Data viewed	Data or settings changed	Systems disrupted	SIEM login anomalies, access logs

8.6 Worked End-to-End Example

Scenario: a finance employee receives a phishing e-mail and the attacker obtains the employee's credentials. (Stages 1-2) The organization has already classified the finance database as a critical asset where Confidentiality and Integrity matter most, and it has controls such as multi-factor authentication on only some systems. (Stage 3 - Detection) The SIEM raises an alert for a login from an unusual country followed by a large data transfer. (Stage 4 - Collection) Analysts collect the e-mail headers, the malicious URL, proxy and VPN logs, and search OSINT and threat feeds for the domain and IP address. (Stage 5 - Processing) The indicators are normalized and de-duplicated, and enrichment shows that the domain was registered three days ago and is linked to a known credential-theft campaign. (Stage 6 - Analysis) Correlation shows the same attacker accessed two other accounts, uses a known set of TTPs and is likely preparing a ransomware deployment; the risk to Confidentiality is rated high and to Availability as likely next. (Stage 7 - Dissemination) The CISO receives a strategic briefing on the campaign and its business risk, the incident-response lead receives an operational report on attacker behaviour and timing, and the engineers receive the IOC list and detection rules for the SIEM and firewall. (Stage 8 - Response) The domain and IP address are blocked, the three accounts are reset, affected hosts are isolated, MFA is enforced everywhere, and staff awareness training is scheduled. These lessons update the security baseline of Stage 1, so the next attack of this kind is stopped earlier.

8.7 Conclusion

The end-to-end concept map shows that network security and threat intelligence are one connected system rather than separate activities. Fundamentals and CIA goals tell the organization what to protect and what an attack would damage; detection finds the threat; collection, processing and analysis turn the evidence into understanding; dissemination makes that understanding available to the right decision-makers; and response converts it into protective action. The feedback loop from response back to the fundamentals ensures that every incident leaves the organization better informed and better defended.